

## Q1 – Dataset Loading and Exploration [10 Marks]

### a) Import Libraries [2 Marks]

Import the necessary Python libraries required for:

- Data manipulation
  - Numerical calculations
  - Data visualization
  - Statistical analysis
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### b) Load the Dataset [2 Marks]

Load `bioinformatics_visualization_practice_dataset.csv` into a Pandas DataFrame.

Display the dataset to confirm that it has been loaded successfully.

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### c) Display the First 5 Records [1 Mark]

Display the first 5 rows of the dataset.

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### d) Display Dataset Information [2 Marks]

Display:

1. Number of rows and columns
  2. Column names
  3. Data types of each column
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### e) Check Missing Values [3 Marks]

Check the dataset for missing values.

Display the number of missing values in each column.

Identify which columns contain missing values.

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## Q2 – Bar Chart: Disease Status [10 Marks]

### Objective

Determine the number of patients belonging to each disease-status category and visualize the result using a bar chart.

### Tasks

#### a) Count Patients [3 Marks]

Find the number of patients in each category of the Disease\_Status column.

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#### b) Create a Bar Chart [5 Marks]

Create a bar chart showing the number of patients in each disease-status category.

The chart must contain:

- Appropriate title
  - X-axis label
  - Y-axis label
  - Clearly visible category names
- 

#### c) Interpretation [2 Marks]

Write one observation based on the bar chart.

Example:

Which disease-status group contains the highest number of patients?

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### Q3 – Clustered Bar Chart: Gene Expression Comparison [12 Marks]

#### Objective

Compare the expression levels of three genes across different disease-status groups.

Use the following columns:

- Gene\_A\_Expression
- Gene\_B\_Expression
- Gene\_C\_Expression
- Disease\_Status

#### Tasks

##### a) Calculate Mean Gene Expression [4 Marks]

Calculate the mean expression level of:

- Gene A
- Gene B
- Gene C

for each Disease\_Status group.

Present the calculated results in a suitable table.

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##### b) Create a Clustered Bar Chart [6 Marks]

Create a clustered bar chart comparing the mean expression levels of Gene A, Gene B, and Gene C across the disease-status groups.

The chart should contain:

- Appropriate title
- X-axis label
- Y-axis label
- Legend identifying Gene A, Gene B, and Gene C
- Clearly labelled disease-status groups

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c) Interpretation [2 Marks]

Based on the clustered bar chart:

1. Which disease-status group generally shows higher gene expression?
  2. Mention one important observation from the chart.
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Q4 – Box Plot: Gene A Expression [10 Marks]

Objective

Analyze the distribution of Gene A expression across different disease-status groups.

Use:

Gene\_A\_Expression

and

Disease\_Status

Tasks

a) Create a Box Plot [6 Marks]

Create a box plot showing Gene\_A\_Expression for each Disease\_Status group.

Include:

- Appropriate title
  - X-axis label
  - Y-axis label
  - Clearly labelled disease-status groups
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b) Compare Median Values [2 Marks]

Examine the box plot and identify:

Which disease-status group has the highest median Gene A expression?

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c) Identify Outliers [2 Marks]

Examine the box plot and determine whether any potential outliers are present.

Briefly explain your observation.

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Q5 – Violin Plot: Gene B Expression [10 Marks]

Objective

Study the distribution of Gene B expression across disease-status groups using a violin plot.

Use:

Gene\_B\_Expression

and

Disease\_Status

Tasks

a) Create a Violin Plot [6 Marks]

Create a violin plot showing Gene B expression for each disease-status group.

Include:

- Appropriate title
  - X-axis label
  - Y-axis label
  - Clearly labelled groups
- 

b) Compare Gene B Expression [2 Marks]

Identify the disease-status group that generally shows higher Gene B expression.

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c) Interpret the Violin Plot [2 Marks]

Explain what the width of a violin represents in a violin plot.

Also mention what you can learn about the distribution of Gene B expression from the plot.

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## Q6 – Heatmap: Correlation Analysis [12 Marks]

### Objective

Analyze the relationships between clinical measurements and gene-expression variables using a correlation heatmap.

Use the following variables:

- Age
- Gene\_A\_Expression
- Gene\_B\_Expression
- Gene\_C\_Expression
- Blood\_Pressure
- Glucose

### Tasks

#### a) Create a Correlation Matrix [4 Marks]

Calculate the correlation matrix for the six selected variables.

Display the correlation values.

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#### b) Create a Heatmap [6 Marks]

Create a correlation heatmap using the calculated correlation matrix.

The heatmap should:

- Display correlation values
- Have appropriate labels
- Have an appropriate title
- Use a suitable colour scale

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c) Interpretation [2 Marks]

Identify the strongest positive relationship between any two variables.

State the two variables and briefly explain the relationship.

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#### Q7 – Volcano Plot: Differential Gene Expression [14 Marks]

Objective

Compare gene-expression levels between Disease and Healthy patients using a volcano plot.

Use the following genes:

- Gene\_A\_Expression
- Gene\_B\_Expression
- Gene\_C\_Expression

Compare:

Disease vs Healthy

Tasks

a) Calculate Mean Expression [2 Marks]

For each gene, calculate:

1. Mean expression in the Disease group
  2. Mean expression in the Healthy group
-

b) Calculate Log2 Fold Change [4 Marks]

Calculate the log2 fold change for each gene using:

$$\text{log2 Fold Change} = \log_2(\text{Mean Disease Expression} / \text{Mean Healthy Expression})$$

Calculate the value separately for:

- Gene A
  - Gene B
  - Gene C
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c) Calculate Statistical Significance [3 Marks]

Perform a two-sided independent t-test comparing the Disease and Healthy groups for each gene.

Record the p-value for:

- Gene A
  - Gene B
  - Gene C
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d) Create a Volcano Plot [4 Marks]

Create a volcano plot using:

- X-axis: Log2 Fold Change
- Y-axis:  $-\log_{10}(\text{p-value})$

Plot:

- Gene A
- Gene B
- Gene C

Label each gene on the plot.

Use the following thresholds:

- $|\log_2\text{FC}| > 0.5$
  - $p < 0.05$
-

e) Interpretation [1 Mark]

Identify which genes satisfy both of the following criteria:

- $|\log_2FC| > 0.5$
- $p < 0.05$

Note: This is an illustrative teaching exercise for learning volcano plots and statistical visualization. It is not a formal biological differential-expression experiment.

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Q8 – UpSet Plot: Mutation Intersections [12 Marks]

Objective

Analyze the occurrence and combinations of mutations using an UpSet plot.

Use the following columns:

- Mutation\_A
- Mutation\_B
- Mutation\_C

Tasks

a) Prepare Mutation Data [3 Marks]

Convert the mutation information into a suitable Boolean/set representation for creating an UpSet plot.

For example:

- Yes  $\rightarrow$  True
  - No  $\rightarrow$  False
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b) Create an UpSet Plot [5 Marks]

Create an UpSet plot showing the intersections between:

- Mutation A
- Mutation B
- Mutation C

If the upsetplot library is not installed, install it before creating the plot.

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c) Identify the Most Common Combination [2 Marks]

From the UpSet plot, identify the most common mutation combination/intersection.

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d) Interpretation [2 Marks]

Explain why an UpSet plot can be more useful than a Venn diagram when analyzing multiple biological sets or mutation combinations.